

Lab Exercise 1: Triggering Nested Loop Join

1	EXPLAIN (ANALYZE , BUFFERS)
2	SELECT c.first_name , c.last_name , r.rental_date
3	FROM customer c
4	JOIN rental r ON c.customer_id = r.customer_id
5	WHERE c.customer_id = 1 ;

Data Output	Messages	Notifications
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Showing rows: 1 to 11 Page No: 1 of 1

	QUERY PLAN text	
1	Nested Loop (cost=0.28..359.16 rows=32 width=21) (actual time=0.133..2.151 rows=32 loops=1)	
2	Buffers: shared hit=153	
3	-> Index Scan using customer_pkey on customer c (cost=0.28..8.29 rows=1 width=17) (actual time=0.086..0.090 rows=1 loo...	
4	Index Cond: (customer_id = 1)	
5	Buffers: shared hit=3	
6	-> Seq Scan on rental r (cost=0.00..350.55 rows=32 width=10) (actual time=0.041..2.048 rows=32 loops=1)	
7	Filter: (customer_id = 1)	
8	Rows Removed by Filter: 16012	
9	Buffers: shared hit=150	
10	Planning Time: 0.497 ms	
11	Execution Time: 2.230 ms	

Identify the Roles: Look at the plan tree. Which table is the Outer (top) and which is the Inner (bottom)?

→ customer is Outer, rental is Inner.

Why Nested Loop? How many rows did the customer table return? Why does this make Hash Join "too expensive"?

→ Because Outer table is very small. If we use Hash Join, we will cost more RAM to initial hash table.

Efficiency: Look at loops in the Inner Table node. Why is it 1?

→ Because the outer scan only 1, the inner scan only execute 1.